

Slides 5.10: Meet the Machine: Parts, Safety, Threading, and Paper Drills

Slide outline for Lesson 5.10, three days, twenty slides. Slides 1 to 8 are Day 1 (parts, rules, your machine). Slides 9 to 14 are Day 2 (threading and the license). Slides 15 to 20 are Day 3 (paper drills). No fabric appears on any slide on purpose; fabric starts in Lesson 5.11.

Slide 1 (Day 1): Three seconds against twenty minutes

- Lesson 5.10, day one of three
- This machine sews in three seconds what your hands did in twenty minutes
- The trade: it does not know your finger is there
- These three days are how you earn the pedal Image: a machine-sewn seam and a hand-backstitched seam side by side, both four inches long.

Slide 2: Do now, guess the seven parts

- Seven parts are numbered on the diagram
- Write what you think each one does. Guess.
- Star the one you think is the most dangerous Image: the numbered machine diagram from Handout 05.10, no labels.

Slide 3: The seven parts

- 1 Needle: up and down, does not care what is under it
- 2 Presser foot: holds the fabric down
- 3 Feed dogs: pull the fabric through, so you guide, never push
- 4 Bobbin: the bottom thread, under the plate
- 5 Take-up lever: the arm the top thread rides
- 6 Hand wheel: one stitch at a time, always toward you
- 7 Foot pedal: the gas Image: the same diagram, labeled.

Slide 4: The most dangerous part

- Most of you starred the needle. Fair.
- The better answer: the pedal
- The pedal moves the needle while a hand is in the wrong place Image: a foot resting beside (not on) a machine pedal.

Slide 5: The four rules

1. Hands to the sides of the needle, never in front. Guide, do not push.
2. One foot on the pedal, the other flat on the floor
3. Power off before changing a needle or reaching under the plate
4. Hair back, sleeves clear, cords taped, nothing loose near the wheel

- The habit that is not a rule: hand wheel toward you, one stitch, look Image: two hands flat on fabric, one on each side of a presser foot, needle centered between them.

Slide 6: Your machine, all unit

- The number on your machine is yours until the unit ends
- Same student, same machine, every day
- You will learn its habits. You will notice when something changed.
- Your threading buddy is the student at the next number Image: a numbered card taped to a sewing machine base, reading 4.

Slide 7: Close-down, and the two hard rules

- Every machine day ends the same way: presser foot down, power off, chair in
- A broken needle is reported and handed over, both pieces, flat on your palm
- Never the trash. Never the floor.
- A jammed machine gets the red doctor tag: tag it, tell me, move to a spare or the hand station, keep working Image: a red MACHINE DOCTOR tag hanging from a machine's arm.

Slide 8 (Day 1 close): Prove it

- Run the close-down routine on your machine, out loud
- Exit line at the door: one part and what it does Image: a row of machines, feet down, a pushed-in chair at each.

Slide 9 (Day 2): Do now, the path from memory

- Write the top threading path in order, as far as you remember
- Do not look at the handout yet Image: a spool of thread alone on a table.

Slide 10: Threading the top, six steps

1. Spool on the pin
2. Through the first guide
3. Down and around the tension dial
4. Up through the take-up lever, back to front
5. Down through the lower guides
6. Through the needle, front to back Image: the threading path drawn as a single line over the machine diagram, numbered.

Slide 11: The bobbin

- Drop the wound bobbin in the case the way the arrow shows
- Thread through the slot, close the plate
- Hold the top thread, hand wheel one full turn toward you
- Pull the loop up, both threads back under the foot Image: a bobbin case with an arrow showing thread direction, and a loop of bobbin thread coming up through the needle plate.

Slide 12: The manual outranks me

- Every machine is a little different
- The manual at your machine is the authority
- When the diagram and your machine disagree, the manual wins Image: a machine manual open in its binder beside a machine.

Slide 13: The threading license

- Thread the top and load the bobbin at your own machine
- Your buddy checks every step against the guide and initials
- I check the tension and the bobbin, and initial
- Two initials make a license: from then on you sit at a machine you threaded Image: the license line from Handout 05.10 with two sets of initials.

Slide 14 (Day 2 close): Unthread it

- Finished early? Unthread the whole machine and thread it again from nothing
- That is the real test
- Close-down routine, then the exit line: the step you will forget tomorrow Image: a machine with thread removed, spool sitting beside it.

Slide 15 (Day 3): Do now, rules from memory

- Write the four rules without looking
- Then write the close-down routine Image: blank lines numbered 1 to 4, then 1 to 3.

Slide 16: Paper, no thread

- Today the machine has no thread at all
- The needle punches holes, and the holes are your record
- Hold the paper up to the light: every wobble shows
- This is how factories train, and it costs nothing Image: a sheet of paper with a drawn line and a row of needle holes running almost exactly along it.

Slide 17: The three sheets

- Sheet 1: the straight line, three passes
- Sheet 2: the curve, slow and steady, small steering
- Sheet 3: the square, with a pivot at each corner Image: the three drill sheets fanned out.

Slide 18: The pivot

- Stop with the needle DOWN in the paper
- Foot UP
- PIVOT the paper around the needle
- Foot DOWN, keep going
- Say it as a chant: needle down, foot up, pivot, foot down Image: a square drill sheet mid-turn, needle down at the corner, presser foot lifted.

Slide 19: What I am checking

- Hands to the sides
- Steady speed
- Holes on the line
- Cannot keep hands to the sides? You drive with the hand wheel until you can. That is a technique, not a penalty. Image: the three-box clipboard checklist.

Slide 20 (Day 3 close): Trade and mark

- Trade your square with your buddy
- Mark their straightest two inches and their wobbliest two inches
- Square sheets come to me: they are the record that the license is real
- Close-down routine. Tomorrow, fabric. Image: two drill squares side by side, two sections circled on each.